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Introduction

Child Mortality is a strong and widely used indicator of population health and development. High child mortality rates could signify some underlying problems in a society, such as a lack of clean drinking water and sanitation, poor living conditions, poor nutrition, and access to health care. There has been substantial research on human capital that reveals a correlation between maternal education and child survival and mortality rates.

In a paper studying the relationship between a mother's education and child growth and nutrition in the first two years of life, Rezaeizadeh et al. found a significant relationship between the two variables. The paper measured child health using z-scores for weight-for-age and height-for-age, which are similar to the health measures discussed in class. They found that children of more educated mothers tended to be healthier. Since a child's health in the first two years after birth is closely related to survival, this study supports the idea that maternal education may also be related to lower child mortality. However, the study also found that the relationship was stronger in lower and middle-income countries, with almost no correlation in higher-income countries. Since Peru is considered a middle-income country, we wanted to test whether the relationship between mothers' education and child mortality would still appear in lower-resource areas, such as rural areas.

Economic research also supports this relationship. Breierova and Duflo (2004) study the relationship between education, fertility, and child mortality using a school construction program in Indonesia. Their paper finds that parental education is associated with lower child mortality, supporting the theory that education can improve child outcomes through better health knowledge, decision-making, and investments in children.

In this paper, we examine the relationship between maternal literacy and child mortality in Peru using household-level census data. We use literacy as a proxy for maternal education due to data availability. Additionally, literacy as a substitute for education works, as it typically indicates that education was achieved at some point in life. Furthermore, to account for hidden-variable bias, we are including variables for house and car ownership and for internet access in our regression. For these variables, we predict they will negatively impact child mortality.

We hypothesize that children born to more educated mothers in rural areas of Peru will have lower mortality rates than those born to less educated mothers. We predict that the negative relationship between maternal education and child mortality will hold even after accounting for additional household and regional factors, and we expect the effect to be much stronger in rural than in urban areas, as rural areas have fewer resources and less access to healthcare.

Empirical Model

Summary Statistics (Table 1)

Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Child Mortality Rate	48,285	0.034	0.116	0.000	1.000
Literate	63,605	0.910	0.286	0	1
Rural	63,605	0.154	0.361	0	1
Owns Home	63,605	0.211	0.408	0	1
Owns Car	62,004	0.150	0.357	0	1
Internet Access	62,004	0.667	0.471	0	1
Age	63,605	42.301	17.365	18	100

To test our hypothesis, we estimate the following OLS regression model with **Equation 1:**

chmortri = β0 + β1liti + β2rurali + β3ownershipi + β4autosi + β5interneti + β6agei + εi

→ Where chmortri is the child mortality rate for mother i, which is defined as the proportion of children born who did not survive. The model is estimated using (PERWT) to ensure estimates are nationally representative.

Variable Description:

lit - Indicator equals 1 if the mother is literate, 0 otherwise. This is a primary variable of interest, serving as a proxy for maternal education and human capital.

rural - Indicator equals 1 if the household is in a rural area. Captures the difference in access to health care and other resources between rural and urban areas.

ownership - Indicator equals 1 if the household owns its home. This is included as a control for household wealth and stability.

autos - Indicator equals 1 if the household owns at least one automobile. This can serve as an additional proxy of household wealth and physical access to healthcare.

internet - Indicator equals 1 if the household has internet access. This is included as a control for access to information (whether that is finding health or education information, or possibly booking a doctor's appointment)

age - Mother's age in years. This is included as a control variable, as older mothers had children in earlier periods when healthcare outcomes were potentially worse.

Predictions:

- Our hypothesis predicts that **$\beta 1 < 0$** (a negative lit coefficient), meaning that literate mothers, who have greater health knowledge and human capital, should have lower child mortality rates.
- We also predict **$\beta 2 > 0$** (a positive rural coefficient), as households in rural areas face greater barriers to accessing healthcare, leading to higher mortality rates.
- For wealth proxies, we predict **$\beta 3 < 0$ and $\beta 4 < 0$** (negative ownership and auto coefficient), as wealthier households have better access and can afford healthcare, which will decrease child mortality.
- For internet, we predict **$\beta 5 < 0$** (a negative internet coefficient), as households with internet access have greater access to health information, which we expect will lead to lower child mortality.
- For age, **$\beta 6$** is somewhat ambiguous. Older mothers may have more resources and experience, but they had children earlier, when healthcare was worse. This means the coefficient could potentially be either positive or negative.

Empirical Model (Table 2)

```
##
## OLS Regression: Determinants of Child Mortality
## ================================================
##                          Dependent variable:
##                      ---------------------------
##                            Child Mortality Rate
## ------------------------------------------------
## Literate                        -0.044***
##                                  (0.002)
##
## Rural                           0.015***
##                                  (0.001)
##
## Owns Home                        0.0001
##                                  (0.001)
##
## Owns Car                        -0.007***
##                                  (0.002)
##
## Internet Access                 0.013***
##                                  (0.001)
##
## Age                             0.002***
##                                 (0.00003)
##
## Constant                        -0.009***
##                                  (0.003)
##
## ------------------------------------------------
## Observations                     47,973
## R2                                0.092
## Adjusted R2                       0.091
## Residual Std. Error      1.382 (df = 47966)
## F Statistic          806.124*** (df = 6; 47966)
## ================================================
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Data

We used data gathered in Peru in 2017 to examine whether there is a correlation between a mother's education and child mortality in rural Peru (Ruggles et al. 2026). In total, 8 variables were filtered: CHBORN, CHSURV, URBAN, OWNERSHIP, LIT, AGE, AUTOS, and INTERNET.

For the variables CHBORN, CHSURV, AGE, LIT, INTERNET, and AUTOS, we filtered to remove any data points labeled as Unknown, Not in Universe, Not Reported, or Missing. For the remaining variables, we filtered them to include only specific qualities in our regression model. For URBAN, we filtered to include only rural households and renamed the variable to "rural". For the OWNERSHIP variable, it was renamed to "ownership" and filtered to include only families who owned their houses. Additionally, in this filter, we created the child mortality variable, chmortr, using the filtered versions of CHBORN and CHSURV (CHBORN_clean, CHSURV_clean).

Results

Summary statistics for the variables used in this analysis are shown in Table 1. The average child mortality rate in the sample is 0.034, meaning that, on average, about 3.4% of children born to mothers in the sample did not survive. The sample has a high overall literacy rate, with about 91% of mothers classified as literate. About 15.4% of households are located in rural areas, 21.1% own their homes, and 15.0% own at least one automobile. Internet access is more common, with about 66.7% of households having it. The average mother in the sample is about 42 years old.

The regression results are shown in Table 2. The coefficient on literacy is negative and statistically significant at the 1% level. This means that literate mothers are associated with a child mortality rate that is about 4.4 percentage points lower than the child mortality rate for illiterate mothers, holding the other variables in the model constant. This result is consistent with the empirical model's prediction that maternal literacy, as a proxy for education and human capital, would be negatively associated with child mortality.

The coefficient on rural residence is positive and statistically significant at the 1% level. Rural households are associated with a child mortality rate that is about 1.5 percentage points higher than urban households, holding the other variables constant. This finding is also

consistent with the prediction that rural households may face greater barriers to accessing healthcare, transportation, and other resources important for child survival.

For the household wealth and access variables, the results are mixed. Car ownership has a negative and statistically significant coefficient, indicating that households with at least one car are associated with a child mortality rate about 0.7 percentage points lower than households without a car. This supports the prediction that access to transportation may be connected to better access to healthcare and other resources. Home ownership, however, is not statistically significant, indicating that the regression does not provide strong evidence of a relationship between home ownership and child mortality in this sample.

The coefficient on internet access is positive and statistically significant, contrary to the original prediction. Households with internet access are associated with a child mortality rate about 1.3 percentage points higher than that of households without internet access, holding other variables constant. This result may reflect omitted variable bias or differences between households that are not fully captured by the model. For example, internet access may be correlated with location, reporting differences, or other household characteristics that are not directly included in the regression.

Finally, the coefficient on age is positive and statistically significant. A one-year increase in the mother's age is associated with a 0.2 percentage-point increase in the child mortality rate, holding other variables constant. This may be because older mothers have children earlier in life, when healthcare access, sanitation, and child survival conditions were worse. Overall, the repression results largely support the paper's main hypothesis: maternal literacy is negatively associated with child mortality, whereas rural residence is positively associated with it.

Conclusion

Our findings align with the theoretical prediction originally made in this paper: there is a statistically significant correlation between a mother's education and the reduction in child mortality in rural Peru. We were able to prove this claim using our OLS regression table, which showed that mothers' literacy, used as a proxy for mothers' education, was associated with a 4.4% decrease in child mortality at the $p<0.01$ level.

We continued to test variables such as home and car ownership and internet access, as we predicted that these variables could impact child mortality without impacting mothers' education.

Of these variables, we were partially correct, as we predicted that all would lower child mortality. Our results showed that home ownership affected child mortality, but not statistically significantly; internet access was significant, and increased child mortality; and car ownership was also significant and decreased child mortality.

In a broader context, our findings support the idea that many families in rural Peru may still face barriers to healthcare and educational opportunities, which contribute to higher child mortality. For example, access to a car may not leave as significant an impact on child mortality in an urban area as it would in a rural area because urban spaces are typically closer to hospitals or have access to public transportation or first responders. Additionally, since urban spaces have greater access to various forms of healthcare, it may be less detrimental for children if their mothers aren't educated and have limited health knowledge, as these families have access to hospitals and pharmacies. Due to these differences, transportation and mothers' education become significantly more important in rural spaces. So, policies aimed at improving access to education and expanding healthcare infrastructure in rural communities may help reduce child mortality rates.

For our data, we identified 2 possible limitations, the first being a measurement error bias, as literacy is not a completely accurate measure of mothers' education. Although it serves as a good substitute, since literacy implies that education was attained at some point, it is not the perfect replacement. This is because some women can achieve literacy outside formal education through family or community members willing to teach them to read. As a result, our data may show higher education rates among mothers because we are misclassifying some literacy rates as education.

Secondly, another possible limitation of our data is the hidden-variable bias associated with our internet variable. The internet variable was included because we believed that internet access improved access to health knowledge, thereby reducing child mortality. Since we found that internet access actually increased child mortality in our OLS model, we theorized that there may be a possible hidden variable bias. One possible explanation is that internet access may be correlated with certain regional characteristics that were not included in the model. For example, some regions with greater internet access may have worse sanitation and access to clean water, which increases child mortality. So, if these factors are omitted from the regression, the results

would suggest that increased internet access is associated with higher child mortality, leading to a case of hidden variable bias.

In conclusion, our research question asked whether mothers' education reduces child mortality rates in rural Peru. We hypothesized that child mortality would decrease as mothers' education increased, and our OLS regression results suggested we were correct. Additionally, we found evidence of other important factors that can impact child mortality but not mothers' education, such as car ownership and internet access. Overall, our findings suggest that areas in rural Peru lack adequate healthcare infrastructure and access to education, which can lead to higher child mortality rates.

References:

Breierova, Lucia, and Esther Duflo. 2004. “The Impact of Education on Fertility and Child Mortality: Do Fathers Really Matter Less Than Mothers?” *NBER Working Paper No. 10513*. National Bureau of Economic Research. https://www.nber.org/papers/w10513

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Ruggles, Steven, Lara L. Cleveland, Rodrigo Lovatón Dávila, Sula Sarkar, Matthew Sobek, Derek Burk, Dan E. Ehrlich, Jane Lee, and Nate Merrill. *IPUMS International: Version 7.7* [dataset]. Minneapolis, MN: IPUMS, 2025. https://doi.org/10.18128/D020.V7.7

---
title: "Mother's Education Correlation"
output: html_document
date: "2026-06-07"

```{r}
library(ipumsr)
library(ggplot2)
library(tidyr)
library(ivreg)
library(dplyr)
library(fixest)
library(modelsummary)
library(stargazer)
library(gt)
library(doBy)
library(coefplot)
```

```{r}
ddi <- read_ipums_ddi("COURSE_MATERIALS/ipumsi_00005.xml")
# ipumsr::ipums_view(ddi)
df_ipums <- read_ipums_micro(ddi)
```

```
df_analysis <-as.data.frame(df_ipums)
head(df_analysis)
```

Data Cleaning

```{r}
df_analysis <- df_analysis %>%
mutate(
CHBORN_clean = ifelse(CHBORN >= 31, NA, CHBORN),
CHSURV_clean = ifelse(CHSURV >= 31, NA, CHSURV),
chmortr = ifelse(CHBORN_clean > 0, 1 - (CHSURV_clean / CHBORN_clean), NA),
rural = ifelse(URBAN == 1, 1, 0),
ownership = ifelse(OWNERSHIP == 2, 1, 0),
lit =  ifelse(LIT == 2, 1,
        ifelse(LIT == 1, 0, NA)),
internet  = ifelse(INTERNET == 1, 1,
            ifelse(INTERNET == 2, 0, NA)),
age = ifelse(AGE < 999, AGE, NA),
autos = ifelse(df_analysis$AUTOS %in% c(1,2,3,4,5,6,7), 1,
        ifelse(df_analysis$AUTOS == 0, 0,
                NA))
)
```

```{r}
df_analysis %>%
  select(chmortr, lit, rural, ownership, autos, internet, age) %>%
  as.data.frame() %>%
  stargazer(type = "text",
          title = "Summary Statistics",
          covariate.labels = c("Child Mortality Rate", "Literate", "Rural",
                        "Owns Home", "Owns Car", "Internet Access", "Age"))
```

Empirical Model

```{r}
m_ols<-lm(chmortr ~ lit + rural + ownership + autos + internet + age,
        data = df_analysis,
        weights = PERWT)
```

```
stargazer(
  m_ols,
  type = "text",
  title = "OLS Regression: Determinants of Child Mortality",
  covariate.labels = c(
    "Literate",
    "Rural",
    "Owns Home",
    "Owns Car",
    "Internet Access",
    "Age"),
  dep.var.labels = "Child Mortality Rate")
```